Model Output Post-Processing
Supporting Document: `IndigoCarbon_US-1_2024_0065`

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1 Summary

This document describes the post-processing of model outputs that are returned from the Soil Metrics API, a process described in Supporting Document `0070` (2023). Post-processing includes several steps, including averaging across baseline threads and making random draws for uncertainty quantification, which are required to estimate emission reductions of SOC and N_2O according to SEP requirements. The results of this post-processing are then given as an input to credit quantification.

2 Post-processing steps

2.1 Inputs: Simulations organized by point, scenario or baseline thread, Monte Carlo draw, and day

The input to the post-processing is a large dataset of output from simulations of DayCent-CR. DayCent-CR is run at each sample point (labeled by an alphanumeric ID called `point_id`) at which soil measurements were collected.

The calibration of DayCent-CR v1.1.0, described in Supporting Document `0046` (2023), estimates a posterior distribution over parameters in DayCent-CR and over parameters of an "inadequacy model" (described below). Draws from this posterior distribution are made during calibration, and they are used in simulations in the carbon program. Thus, at each point at which DayCent-CR is run, and for the project-scenario and for each of the baseline threads, there are $L = 176$ simulations for each of the draws from the posterior distribution.

For the validation report Supporting Document `0046`, 2023, the Soil Metrics API produces output at a **daily** time step. (For previous validation reports, a monthly time step was used to reduce the size of the data.) A daily time-step is used because fluxes of N_2O are highly variable from one day to the next, so it is difficult to aggregate it to a monthly time step and later interpolate it, as was done for previous calibrations of DayCent-CR.

2.2 Aggregating DayCent-CR output at the cultivation cycle level

The first step in post-processing is to aggregate the daily data to the cultivation cycle level. The start and end dates of cultivation cycles (within each management zone) are retrieved from a table called the Zone Attribute Table (ZAT; see Supporting Document `0048` (2023) for details). To retrieve these start and end dates of cultivation cycles at the points at which soil samples are collected and DayCent-CR is run, the Zone Attribute Table is geospatially joined to the sample points, creating a dataset called the Point Attribute Table (PAT). The PAT is then joined to the DayCent-CR output on the `point_id` column. For each simulation, the data is grouped by cultivation cycle, and the daily SOC stock changes and N_2O emissions are summed across days within the cycle. Only cultivation cycles that will be reported in the monitoring report of interest are selected (i.e., cycles reported in previous monitoring reports are filtered out).

2.3 Averaging across baseline threads

The next step in post-processing is to average across baseline threads. For reasons described in Supporting Document `0048` (2023), we use only blended baselines. The DayCent-CR output, aggregated to the cultivation cycle level at each sample point and Monte Carlo draw, is averaged across the baseline threads using an unweighted average, matching the logic in *Soil Enrichment Protocol v1.1* (2021, Table 5.7).

2.4 Random draws for uncertainty quantification

For uncertainty estimation, we used the Monte Carlo approach in Appendix D.2 in the *Soil Enrichment Protocol v1.1* (2021), with $L = 176$ draws from the posterior distribution of both DayCent-CR parameters and variance parameters as described in Supporting Document `0046` (2023) in accordance with the *SEP Model Requirements and Guidance v1.1a* (2022). This section covers inclusion of additional uncertainty (including structural uncertainty) on model prediction captured by the variation in DayCent-CR parameters estimated during model calibration. The Monte Carlo method estimates the uncertainty of emission reductions calculated for baseline- and project-scenario model simulations.

One source of uncertainty arises from the variation in the calibration parameters (parameter uncertainty) among the $L = 176$ draws from the posterior distribution. These calibration parameters are internal to the DayCent-CR model, and they affect phenomena such as the decay rate of SOC pools.

The other source of uncertainty is captured by variation in parameters of the inadequacy model that is added to the output of DayCent-CR to capture the variation in the calibration data. The same form for the inadequacy model is used for both SOC and N_2O. The three noise terms are:

1. a *site effect*, where "site" is interpreted as "field";
2. a *site-by-year effect*, where, site is interpreted as field, and "year" is calendar year;
3. a *residual effect*, which was modeled as an exponential function of time (for more details see Supporting Document `0046` (2023)).

All three noise terms are normally distributed with zero mean and variance determined by four variance parameters (one parameter for the site effect, one parameter for the site-by-year effect, and two parameters for the residual effect). For each of $L = 176$ draws, the three noise terms are drawn independent of each other. The site effect captures spatial correlation; the site-by-year effect captures spatiotemporal correlation; and the residual effect captures the unexplained residual variance during model calibration.

The Soil Enrichment Protocol is not explicit about how to estimate uncertainty of baseline-scenario emissions given that baseline-scenario emission reductions are estimated by averaging across multiple baseline threads. Specifically, in the context of the revised (*Soil Enrichment Protocol v1.1* 2021, Appendix D), the SEP does not specify the order in which one should add random noise terms and compute the average across baseline.

1. **Approach 1**: Compute "average(baseline threads) + noise" (i.e., average emissions reduction across baseline threads first, then add noise); or
2. **Approach 2**: Compute "average(baseline threads + noise)" (i.e., add noise to the emissions reductions in the individual baseline threads, and then average across baseline threads).

Approach 2 has the strange feature that uncertainty decreases as the number of baseline threads increases. Also, Approach 1 is more conservative in that it results in higher uncertainty. It also provides similar variance for both the project and the baseline average. For these reasons, we chose to implement Approach 1.

2.4.1 Inadequacy model for SOC

For SOC, random draws are added to the logarithm of the SOC stocks (after stocks are shifted by a constant, $1 - Y_{\min}$, as described below), and then this quantity is exponentiated to convert it back to SOC stocks. This transformation of the modeled SOC stocks using the natural logarithm (with a data shift) matches the procedure used in calibration (Supporting Document `0046`, 2023, Equation 2). Specifically, the SOC stock is modeled as

$$Y_{\mathrm{SOC},kls}(t) = \exp\left[\log\left(1 - Y_{\min} + M_{\mathrm{SOC},kls}(t)\right) + \gamma_{\mathrm{SOC},i} + \gamma_{\mathrm{SOC},ij} + \epsilon_{\mathrm{SOC},ks}(t)\right] + Y_{\min} - 1 \tag{1}$$

where

- $M_{\mathrm{SOC},kls}(t)$ is the DayCent-CR prediction for the SOC stock at the end of date t at point k, Monte Carlo draw l, and scenario s (the project scenario or the average across the baseline threads);
- $Y_{\mathrm{SOC},kls}(t)$ is the SOC stock with random draws from the inadequacy model added (on the logarithmic scale) at point k, Monte Carlo draw l, and scenario s;
- γ_i is the random draw from the site effect (where i is the field containing point k);
- γ_{ij} is the random draw from the site effect at site i in year j (the year in which time t belongs);
- $\epsilon_{ks}(t)$ is the draw from the residual random effect at point k, scenario s, and time t;
- $Y_{\min}$ is set to the value used in calibration of DayCent-CR, -461. (The smallest value of N_2O flux was -460 g N/ha, and this transformation of $Y \mapsto 1 - Y_{\min} + Y$ avoided applying the logarithm to negative numbers. The same transformation was used for N_2O and for SOC.)

Because the SOC stock is the same in the project and baseline scenario at the additionality start date, the draw from the residual random effect at the additionality start date is the same in the project and baseline scenarios. Also, the draw of the residual random effect at the end of cultivation cycle is set equal to the draw of the residual random effect at the start of the next cycle (since they both represent the same quantity: the residual draw at midnight on the final day of the earlier of those two cultivation cycles).

In calibration, the variance of the residual random effect grows with the time since the initial measurement Supporting Document `0046`, 2023, Equation 3. Here, we use the additionality start date instead of the time of the initial measurement, since that is often earlier; this choice is conservative because the variance grows with time.

2.4.2 Inadequacy model for direct nitrous oxide emissions

The inadequacy model for the flux of direct N_2O emissions has the same form as that of SOC:

$$Y_{kls}^{\mathrm{Direct}N_2O}(t_1, t_2) = \exp\left[\log\left(1 - Y_{\min} + M_{kls}^{\mathrm{Direct}N_2O}(t_1, t_2)\right) + \gamma_i^{\mathrm{Direct}N_2O} + \gamma_{ij}^{\mathrm{Direct}N_2O} + \epsilon_{ks}^{\mathrm{Direct}N_2O}(t_1, t_2)\right] + Y_{\min} - 1 \tag{2}$$

where

- $M_{kls}^{\mathrm{Direct}N_2O}(t_1, t_2)$ is the DayCent-CR prediction of the flux of direct N_2O between dates t_1 and t_2 (inclusive);

- $Y_{kls}^{\text{Direct}N_2O}(t_1, t_2)$ is the DayCent-CR prediction $M_{kls}^{\text{Direct}N_2O}(t_1, t_2)$ with random draws added (on the logarithmic scale);
- $\gamma_i^{\text{Direct}N_2O}$ is the random draw from the site effect (again, i is the field containing point k);
- $\gamma_{ij}^{\text{Direct}N_2O}$ is the random draw from the site-by-year effect;
- $\epsilon_{ks}^{\text{Direct}N_2O}(t_1, t_2)$ is the draw from the residual random effect;
- $Y_{\min}$ is set to the value used in calibration of DayCent-CR, -461. (The smallest value of N_2O flux was -460 g N/ha, and this transformation avoided applying the logarithm to negative numbers.)

We assume one draw is taken from the residual random effect every 217.1151 days, the average duration of the observations of N_2O flux in the calibration data. To implement that for a cultivation cycle of arbitrary duration, we divide the duration of the cycle by 217.1151 days, and we inflate the variance of the residual random effect by that ratio (exploiting the fact that a sum of D independent draws from a normal distribution is equivalent to one draw from a distribution with D-fold greater variance).

2.5 Calculate Emissions Reductions

As the final step to calculate emission reductions, we do the following calculations at each point, Monte Carlo draw, and cultivation cycle.

2.5.1 Emissions reduction of SOC

First, we calculate the temporal change in SOC stocks for the project scenario and for the baseline scenario. Then we subtract the latter from the former to compute the emissions reduction in the SOC pool. Finally, we convert from g C/m^2 to tCO_2e/ha.

2.5.2 Emissions reduction of direct nitrous oxide

First, we convert mass of N_2O-N to mass of N_2O using the molecular mass ratio. Then we apply time-varying values of global warming potential (GWP), as prescribed by the Climate Action Reserve (i.e., use IPCC AR4 values for vintages 2020 and earlier; use IPCC AR5 values for vintages 2021 and later) (Climate Action Reserve, 2022). To do so, we use the year when the cultivation cycle ends to look up the GWP for N_2O to use for that cultivation cycle.

Finally, we compute emissions reduction of direct N_2O by subtracting the flux of N_2O from the soil to the atmosphere in the project scenario from that same flux in the baseline scenario.

3 Validation domain and model quality control (QC)

3.1 Direct nitrous oxide emissions

3.1.1 Sources of nitrous oxide emissions captured by DayCent-CR

We use DayCent-CR to simulate the flux of N_2O emissions from the soil. These predictions of direct N_2O from DayCent-CR supersede the following default equations in the *Soil Enrichment Protocol v1.1* (2021):

- Equation 5.20: Direct Nitrous Oxide Emissions from Fertilizer
- Equation 5.25: Nitrous Oxide Emissions from the Incorporation of All Crop Residues

Note that we do not provide grazing events to DayCent-CR, so DayCent-CR does not supersede the direct N_2O emissions from manure deposition (Equation 5.23).

3.1.2 Validation domain for direct nitrous oxide emissions

The validation domain for direct N_2O is specified by Table 9 in the validation report (Supporting Document `0046`, 2023). To implement that criteria, we exclude management zones that satisfy one or more of the following:

1. Cotton was planted in the management zone (see Supporting Document `0046` (2023, Table 9, column "C3S")).

2. A crop in the Soy CFG was grown in the management zone, and organic amendments was detected as a practice change in the management zone (see Supporting Document `0046` (2023, Table 9, column "C3AN")).

3. The management zone has grown a crop in the perennial alfalfa crop functional group (CFG) and does a practice other than those in the "cropping" category (see Supporting Document `0046` (2023, Table 9, column "C3PN")).

The results of the calculation of the inclusion criteria for the direct N_2O validation domain are shared in the Data Package in `results/direct_n2o/zone/in_n2o_validation_domain.csv`.

Figure 1 depicts the partitioning of area in the project into two parts, inside and outside the N_2O validation domain.

1. Within the validation domain, DayCent-CR is run at the points at which soil samples were collected. The simulations of direct N_2O are aggregated across that region (i.e., the land inside the direct N_2O validation domain) using the sample design, just like for SOC, using the Monte Carlo methods in *Soil Enrichment Protocol v1.1* 2021, Appendix D.2. (Default equations are used at sample points that fail QC checks, as described in Section 3.1.3 below.) The estimate of the total direct N_2O emissions reduction inside the direct N_2O validation domain is shared in the Data Package in `results/direct_n2o_in_validation_domain/`:

    - `issuance/issuance_results_for_data_package.csv` has results across the whole N_2O validation domain
    - `stratum/results_by_stratum_for_data_package.csv` has results by stratum within the N_2O validation domain
    - `monte_carlo/draws_predictive_distribution_of_total_by_monte_carlo_id.csv` has the posterior predictive mean by sample point, for points inside zones that belong to the direct N_2O validation domain
    - `stratum_and_monte_carlo/draws_predictive_distribution_of_total_by_stratum_and_monte_carlo.csv` has the stratum-wide totals by Monte Carlo draw, for zones inside the direct N_2O validation domain

2. Outside the validation, default equations 5.20 and 5.25 are run on every management zone, just like how the other default equations are run on every management zone.

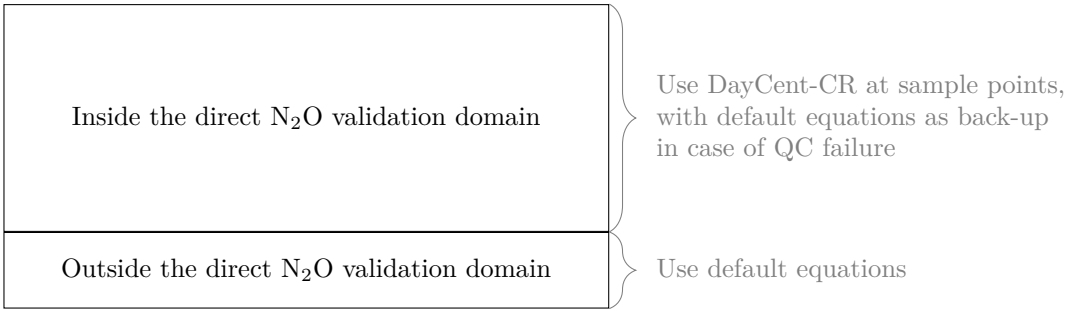


Figure 1: Zones inside the validation domain for direct N_2O use DayCent-CR (pending quality checks described in Figure 2 below), while zones outside the validation domain use default equations run on every management zone.

3.1.3 Quality checks of simulations of direct nitrous oxide emissions

As described in the validation report (Supporting Document `0046`, 2023, Section "Restrictions on application of model") a set of model results that satisfy one of the following criteria was restricted from use in crediting:

1. Mineral soil nitrogen is typically a small proportion of total soil nitrogen (Schulten and Schnitzer, 1997), so modeled N2O emissions will not be used for crediting when modeled mineral nitrogen exceeds 30 percent of total soil nitrogen, which exceeds the pool of even potentially mineralizable nitrogen in most soils (Nendel, Melzer, and Thorburn, 2019).

2. Nitrogen losses to surface water and groundwater by runoff and leaching from cropland is typically between 5 to 50 kilograms per hectare in the United States White et al., 2015. When DayCent-CR predicts nitrogen losses to water that exceed 100 kilograms per hectare, the model will not be applied for N_2O crediting.

3. Nitrogen cycling and transport is sensitive to hydrologic processes (e.g., soil moisture, infiltration). Therefore modeled N2O outcomes will not be applied when all water supplied through precipitation and/or irrigation is lost to evaporation and transpiration (i.e., no predicted runoff or infiltration) (Potter et al., 2006).

When a simulation reaches one of the states 1–3 is reached in at least one Monte Carlo simulation at a sample point, then predictions from default equations 5.20 and 5.25 in the *Soil Enrichment Protocol v1.1* (2021) are used at that sample point, instead of the DayCent-CR prediction of the direct N_2O flux at that sample point. This logic is depicted in the flow chart in Figure 2.

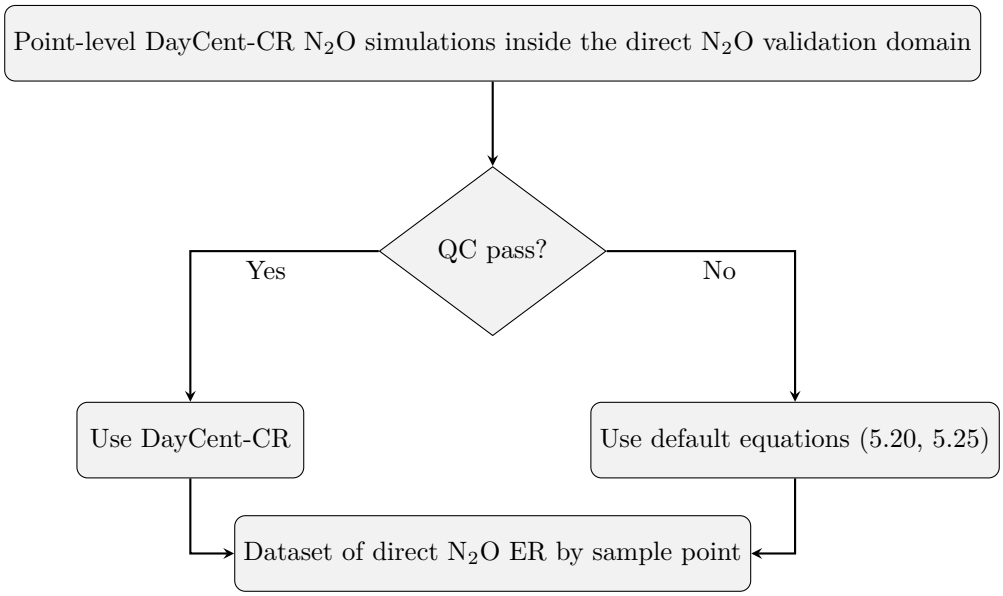


Figure 2: Flow diagram for falling back on default equations at sample points that fail model QC for direct N_2O

The file `results/default_emissions/zone_and_cultivation_cycle/default_equation_results_used.csv` in the Data Package has the results of default equations that were used, either because the zone is outside the direct N_2O validation domain (Figure 1) or because the zone contains a sample point that failed a QC check (Figure 2). For default equations other than 5.20 and 5.25 in the *Soil Enrichment Protocol v1.1* 2021, the equation is run on every management zone, so those columns are not null in this CSV file.

3.2 SOC

3.2.1 Validation domain for SOC

The validation domain for SOC is specified in Supporting Document `0046`, 2023, Table 8. We implemented the inclusion criteria for SOC by excluding from the project all fields that are outside the validation domain for SOC.

3.2.2 Quality checks for SOC

This logic is described in a separate document, `IndigoCarbon_US-1_2024_0075`.

References

[1] Climate Action Reserve. *Use of global warming potential values for all offset protocols.* https://www.climateactionreserve.org/wp-content/uploads/2022/07/2022-GWP-Update-Memo_07.13.22.pdf. Accessed: 2024-06-24. 2022.

[2] Claas Nendel, Dennis Melzer, and Peter J. Thorburn. "The nitrogen nutrition potential of arable soils". In: *Scientific Reports* 9.1 (Apr. 2019), p. 5851. DOI: 10.1038/s41598-019-42274-y. URL: https://doi.org/10.1038/s41598-019-42274-y.

[3] Steven R. Potter et al. *Model Simulation of Soil Loss, Nutrient Loss, and Change in Soil Organic Carbon Associated with Crop Production (Conservation Effects Assessment Project (CEAP))*. Tech. rep. USDA Natural Resources Conservation Service, 2006. URL: https://www.nrcs.usda.gov/publications/ceap-crop-2006-model-simulation-soil-nutrient-full.pdf.

[4] H.-R. Schulten and M. Schnitzer. "The chemistry of soil organic nitrogen: a review". In: *Biology and Fertility of Soils* 26.1 (Nov. 1997), pp. 1–15. DOI: 10.1007/s003740050335.

[5] *SEP Model Requirements and Guidance v1.1a*. Requirements and Guidance for Model Calibration, Validation, Uncertainty, and Verification. Apr. 2022. URL: https://www.climateactionreserve.org/wp-content/uploads/2022/04/SEP_Model_Cal_Val_Guidance_4.2022.pdf.
[6] *Soil Enrichment Protocol v1.1*. May 2021. URL: https://www.climateactionreserve.org/wp-content/uploads/2022/06/Soil-Enrichment-Protocol-V_1.1-final.pdf.
[7] Supporting Document `0046`. *Soil Metrics Type 1 Validation Report*. `IndigoCarbon_US-1_2023_0046`. 2023.
[8] Supporting Document `0048`. *Logic used to construct baseline threads*. `IndigoCarbon_US-1_2023_0048`. 2023.
[9] Supporting Document `0070`. *Running the model*. `IndigoCarbon_US-1_2023_0070`. 2023.
[10] Michael White et al. "Development of Sediment and Nutrient Export Coefficients for U.S. Ecoregions". In: *JAWRA Journal of the American Water Resources Association* 51.3 (2015), pp. 758–775. DOI: https://doi.org/10.1111/jawr.12270.

4 Summary of Changes

Section	Date	Summary of Changes
2.1	2024-06-21	Added description of the input data to post-processing
2.2	2024-06-21	Soil Metrics API now gives daily time step
2.4	2024-06-24	Updated for the inadequacy model used in the new validation report (Supporting Document `0046`, 2023)
2.5.2	2024-06-24	Added details on computing emissions reduction of direct N_2O
3	2024-06-25	Added details on the inclusion criteria for the validation domain and on model QC